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The Research Topics in VR & AR and their Applications

Virtual Reality (VR)

Augmented Reality (AR)

Applications of VR & AR in Practice

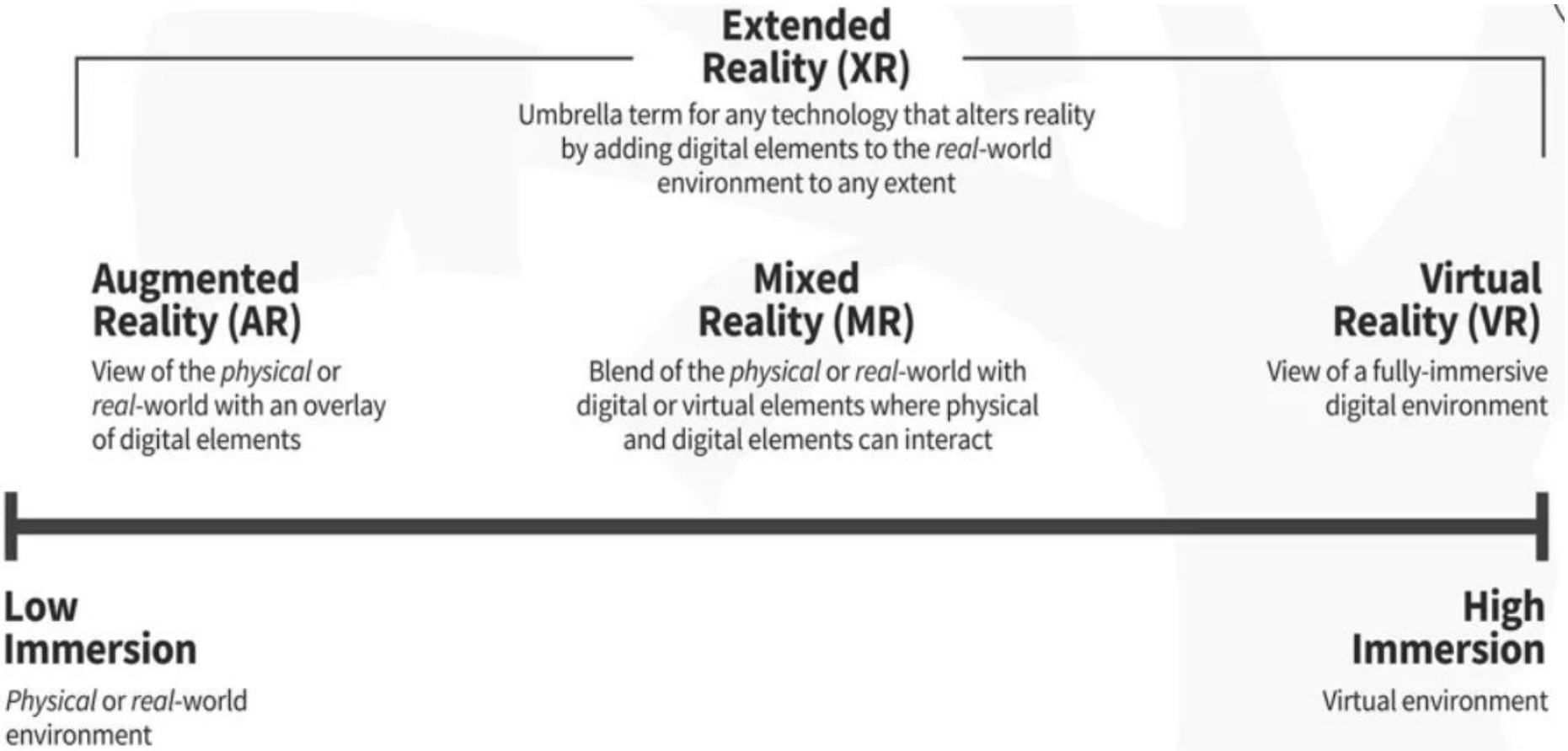
14/4/2025

Virtual Reality (VR) and Augmented Reality (AR)

- Introduction to VR & AR
- Introduction to Unity 3D
- Programing language and IDE
- APIs, Libs and other tools to support
- Devices and Implementation
- Application in practice

Introduction

How do we Compare: VR, AR and MR



Comparison

- VR encompasses immersive experiences and content via a VR headset or HMD (head-mounted display).
- The content is 100% digital and computer-generated
- The current reality is replaced with a new 3D digital environment in which the user is isolated from the real world, but is immersed in the virtual environment

Comparison

Some VR headsets manufacturers include technology giants such as:

- Facebook (through their acquisition of Oculus),
- Google (with a series of Daydream headsets), HTC (VIVE),
- Samsung (Gear VR), and
- Windows (who have named their range of devices Windows Mixed Reality (MR), even though a number is VR-only enabled.)

Comparison

There are three types of VR devices available these are:

- Tethered,
- Stand-alone, and
- Smartphone VR

A phone-based VR headset uses a smartphone for the screen, processor, and sensors in a VR headset. This allows the headset itself to be much cheaper, as the user provide the phone. However, the sensors in a smartphone aren't as good at the ones in a VR headset, and the computer on a smartphone is weak.

Augmented reality (AR)

- Overlays computer-generated content on top of the real world.
- It can superficially interact with the environment in real-time.
- Is primarily experienced via a wearable glass device or through smartphone applications

Augmented reality (AR)

- Augmented content doesn’t recognize or interact with the physical objects within a real-world environment, however, it does enhance the user’s experience
- Some of the categories of AR products are wearable AR glass (smart glass), smartphone AR and other AR headsets

Mixed reality (MR)

Combines several technologies into one wearable device.

- Immersive media,
- spatial computing, and
- hybrid reality intertwine digital content while interacting with a user's real-world environment.

- It allows for digital content to integrate, enrich and interact with the user's real-world environment.
- MR lenses or headsets present an overlay of digital content that interacts with objects in the real world in real-time.
- The products are, in most cases, in the research and development phase, but MR is viewed through transparent wearable glasses

Mixed reality (MR)

- Mixed Reality removes the boundaries between real and virtual worlds using occlusion: the computer-generated objects can be visibly obscured by objects in the physical environment from the user’s point of view
- Some of MR products are Microsoft's hololens, magic leaps and other MR devices

DIFFERENCES BETWEEN VR, AR, AND MR

- VR is content which is 100% digital and can be enjoyed in a fully immersive environment.
- AR overlays digital content on top of the real-world.
- MR is a digital overlay that allows interactive virtual elements to integrate and interact with the real-world environment.

DIFFERENCES

- AR is a variation of the more known concept of VR Technology, which is often defined as “the use of real-time digital computers and other special hardware and software to generate a simulation of an alternate world or environment, which is believable as real or true by the users”.
- VR technology creates an environment in which the user feels and seems to be moving inside a computer-created virtual world in the same way people move inside the natural environment; while immersed in the virtual world, the user cannot perceive the real one which still surrounds him.
- On the contrary, AR allows the user to see the real world, augmenting it with superimposed virtual objects.
- In other words, while VR replaces reality, AR supplements it, creating an environment in which real and virtual objects harmonically coexist.
- AR exploits users’ perceptual-motor skills in the real world, creating a special type of human-machine interaction

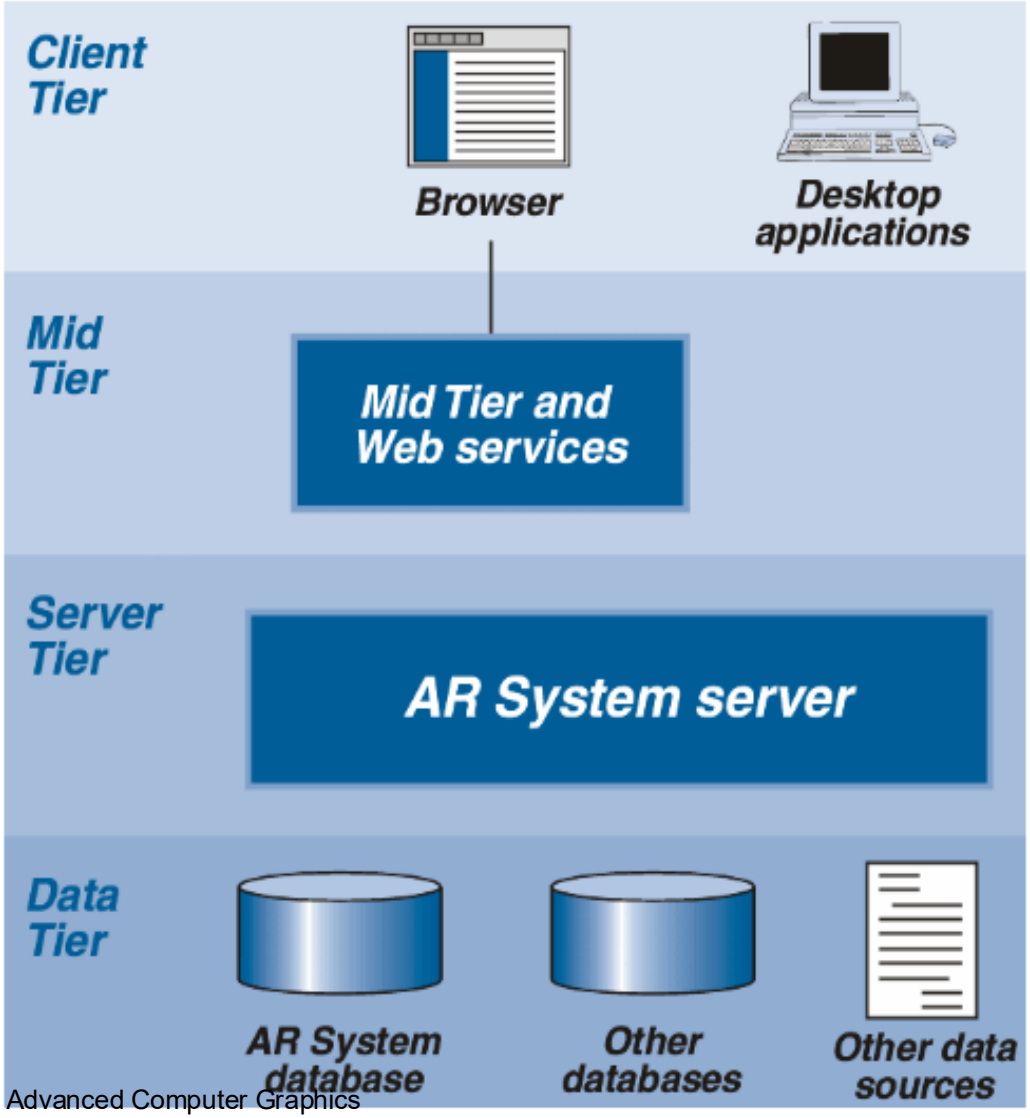
Relation between AR and Telepresence

- In telepresence, the fundamental purpose is to extend the operator's sensory-motor facilities and problem-solving abilities to a remote environment.
- VR aims to achieve the illusion of presence within a computer simulation whereas telepresence aims to achieve the illusion of presence at a remote location.
- AR can be considered a technology between VR and Telepresence.
  - In VR the environment is completely synthetic
  - In Telepresence it is completely real,
  - In AR the user sees the real world augmented with virtual objects.

The architecture of AR systems

- Scene Generator
- Tracking System
- Display

AR Architecture

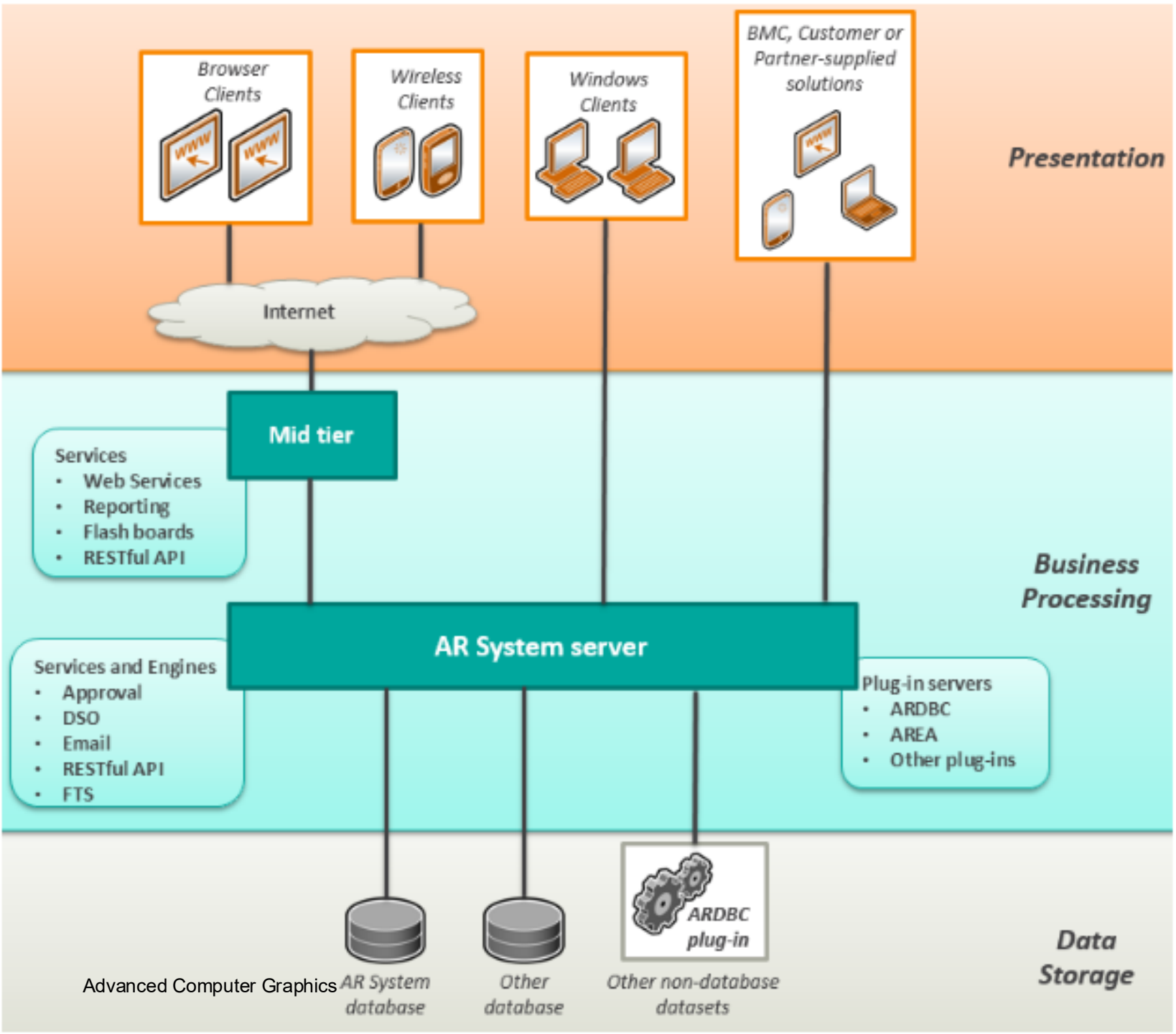


AR Architecture

- AR System clients provide the user interface.
- The Mid Tier makes the user interface available in browsers.
- The AR System server implements the workflow functions, access control, and flow of data into and out of the database.
- The database server acts as a data storage and retrieval engine

Functionally the AR System consists of three environments

- Presentation
- Business Processing
- Data Storage


Browser Clients
Wireless Clients
Windows Clients
BMC, Customer or Partner-supplied solutions
Presentation
Internet
Mid tier
Services
• Web Services
• Reporting
• Flash boards
• RESTful API
Business Processing
AR System server
Services and Engines
• Approval
• DSO
• Email
• RESTful API
• FTS
Plug-in servers
• ARDBC
• AREA
• Other plug-ins
ARDBC plug-in
Data Storage
AR System database
Other database
Other non-database datasets

AR System clients

AR System clients can be broadly divided into

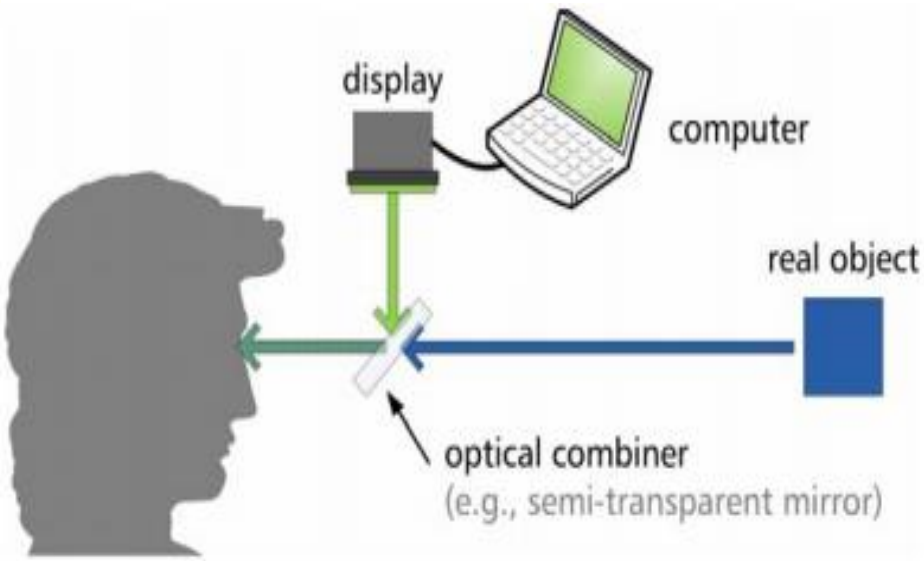
- user client and
- developer clients

Refer: https://docs.bmc.com/docs/ars213/ar-system-architecture-1027564255.html

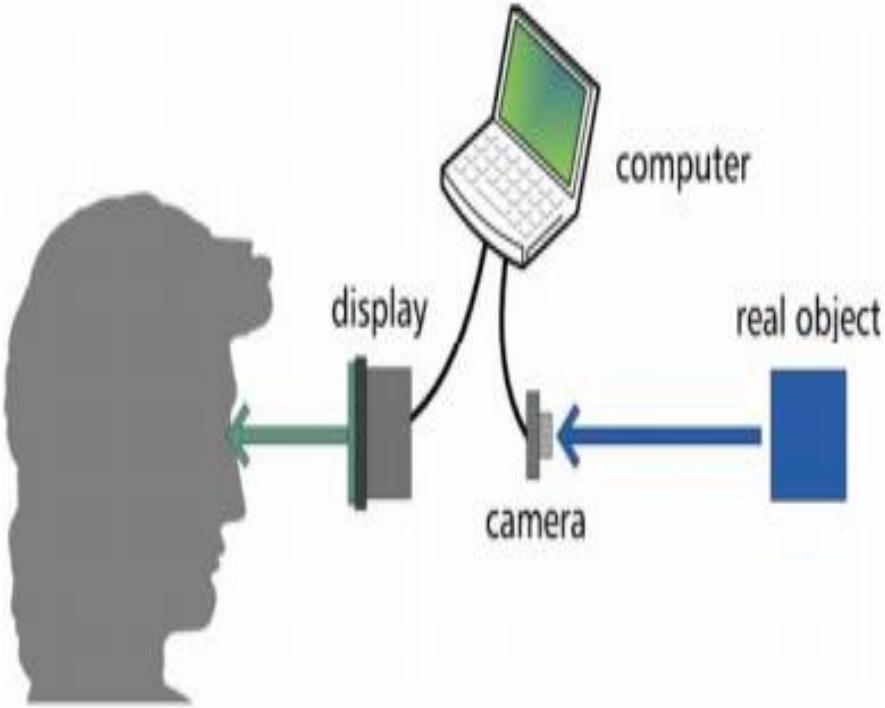
Four major classes of AR distinguished by their display type

- Optical SeeThrough,
- Virtual Retinal Systems,
- Video See-Through,
- Monitor Based AR and Projector Based AR

Optical See-Through HMD



Video See-Through HMD

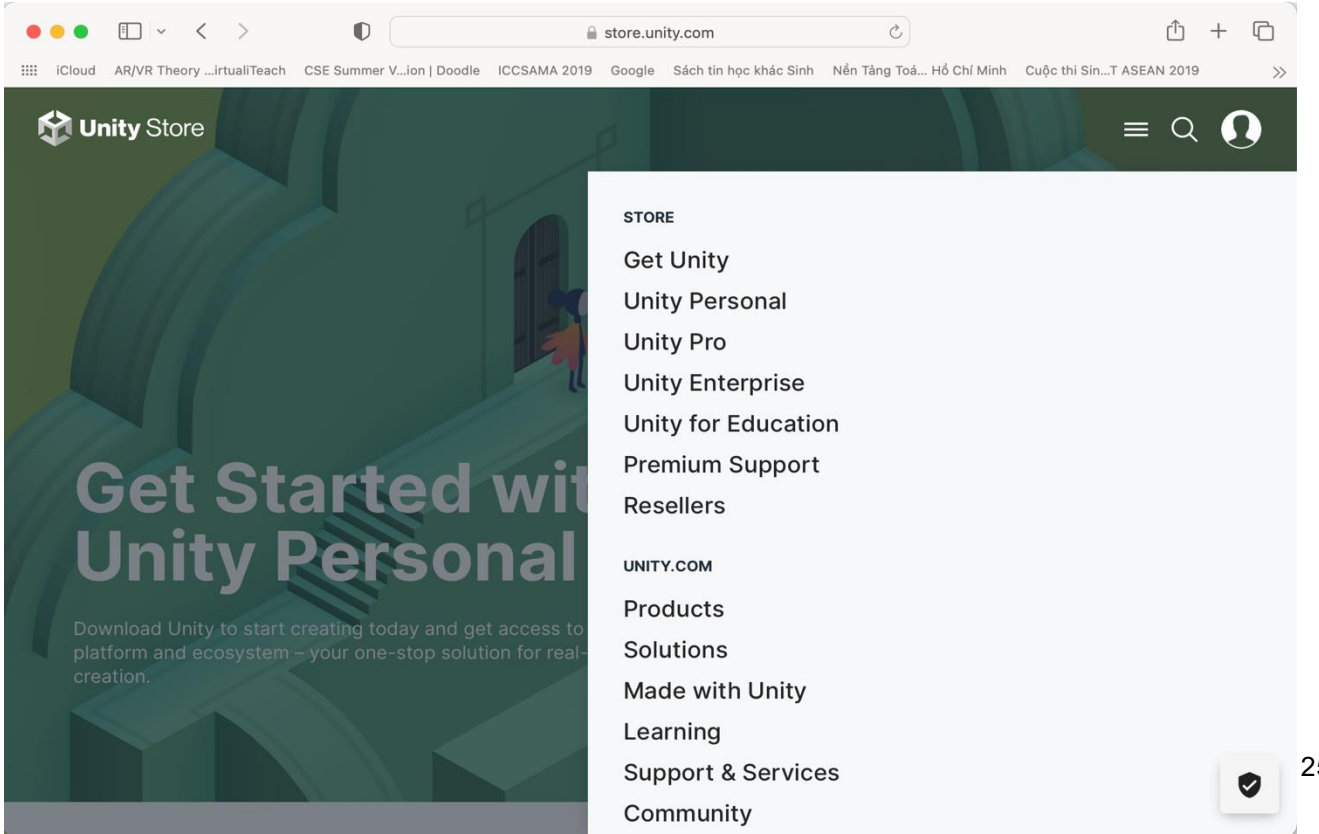


Introduction to Unity 3D

- Unity3D is a widely-used cross-platform game develop system.
- It consists of a game engine and an IDE.
- It can be used to develop games for many different platforms, including PCs, consoles, mobile devices, and deployment on the Web.
- However, this is a complex system, and we will not have time to delve into its many features.
- A good starting point for learning about Unity is to try the many tutorials available on the Unity Tutorial Web page, unity3d.com/learn/tutorials/

Introduction to 3D Unity

- Unity is a cross-platform game development system
- Consists of a game engine and an IDE
- Can be used to develop games and applications for many different AR/VR platforms
- Can be downloaded at: https://store.unity.com/download



Documentation

- Unity User Manual: https://docs.unity3d.com/Manual/index.html
- Scripting API: https://docs.unity3d.com/ScriptReference/index.html
- These pages should become your best friends.
- Also documentation on the OVR Utilities Plugin: https://developer.oculus.com

Unity Official Scripting Videos: These also serve as a good introduction to C#.

- Beginner Scripting Playlist: https://www.youtube.com/watch?v=Z0Z7xc18CcA&list=PLX2vGYjWbI0S9-X2Q021GUtolTqbUBB9B
- Intermediate Scripting Playlist: https://www.youtube.com/watch?v=HzIqrlSbjjU&list=PLX2vGYjWbI0S8YpPPKKvXZayCjkKj4bUP

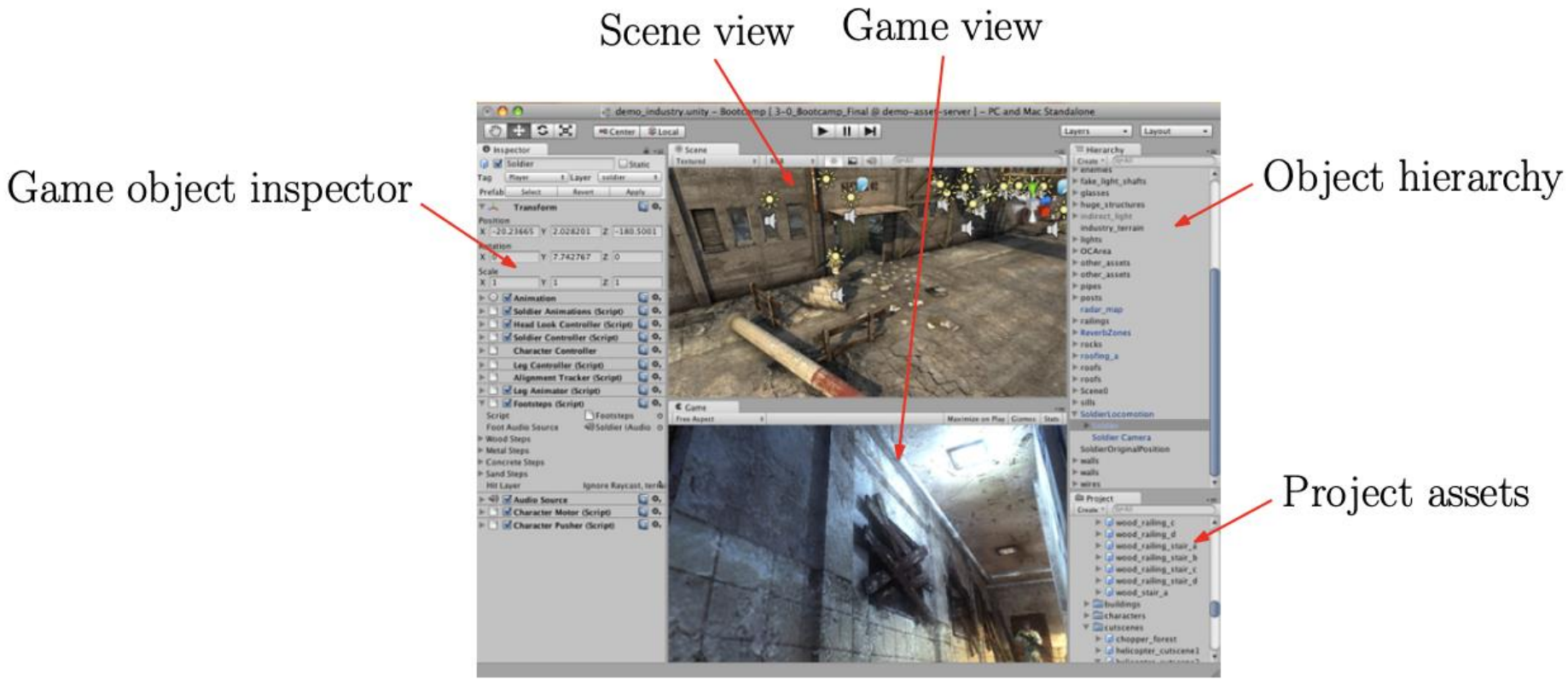
Unity Basic Concepts

- Project: the project contains all the elements that makes up the game, including models, assets, scripts, scenes, and so on.
- Scenes: a scene contains a collection of game objects that constitute the world that the player sees at any time. A game generally will contain many scenes.
- Packages: a package is an aggregation of game objects and their associated meta-data (the objects: models, scripts, materials, etc.)
- Prefabs: a prefab is a template for grouping various assets under a single header. It is used for creating multiple instances of a common object.

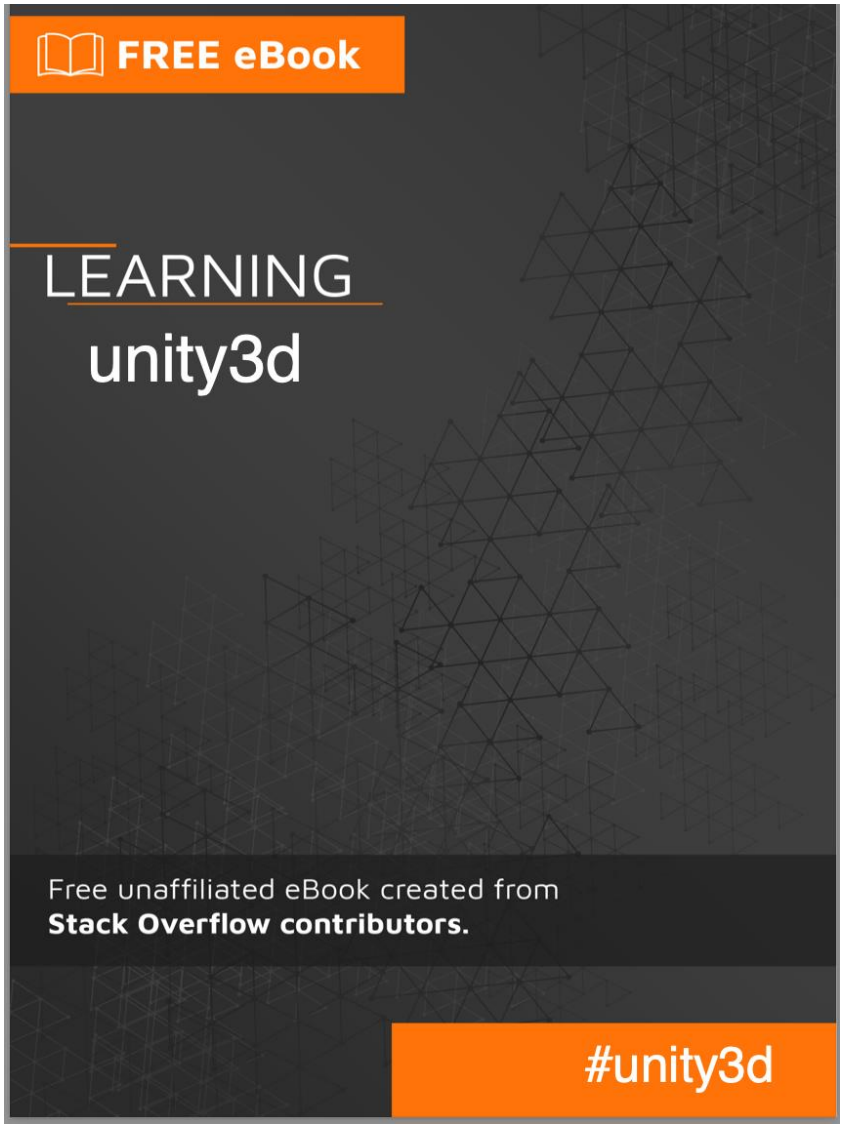
Unity Basic Concepts

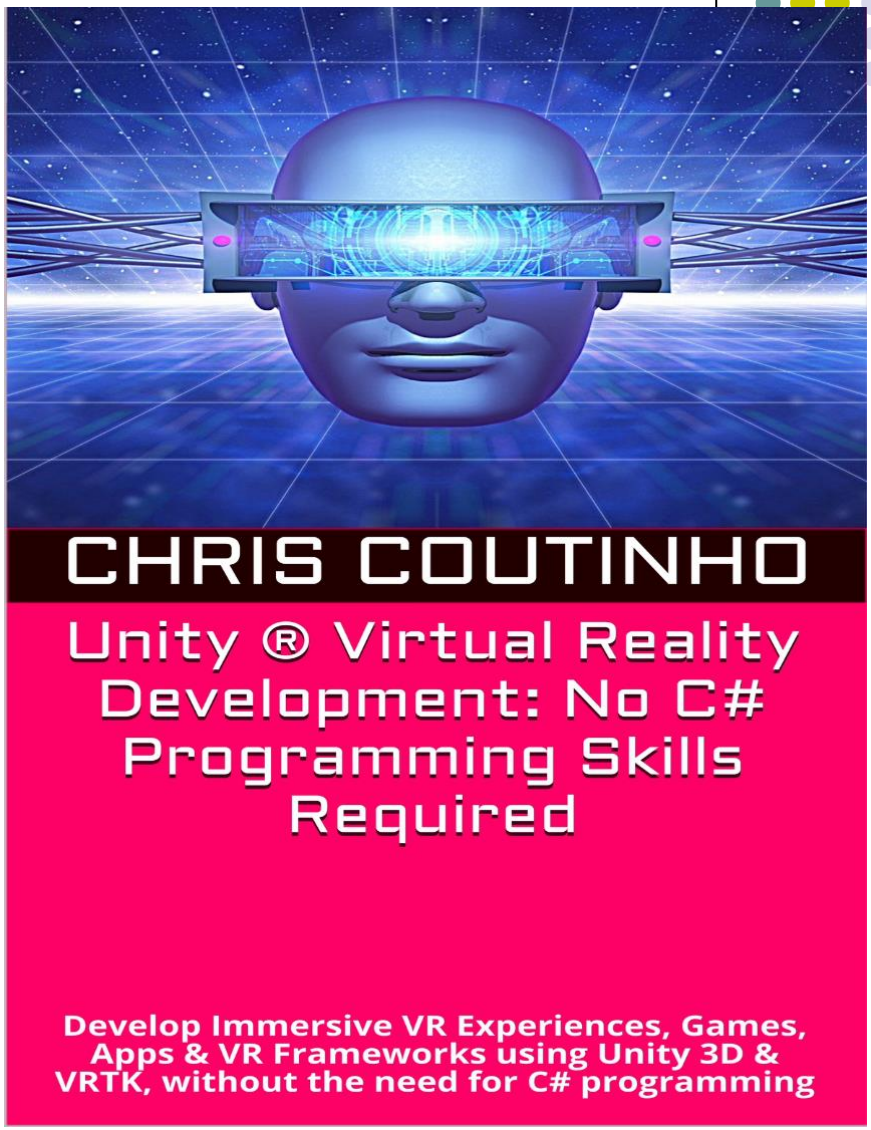
- Game Objects: the game objects are all the “things” that constitute your scene. Game objects not only include concrete objects, but also other elements that reside in space such as light sources, audio sources, and cameras
- Components: each game object is defined by a collection of associated elements (mesh filter, rigid body, collider, surface materials scripts)
- Assets: an asset is any resource that will be used as part of an object’s component: meshes (defining the shapes of objects), materials (defining shapes), physics materials (defining physical properties like friction), and scripts (defining behaviors)
- Scripts: A script is a chunk of code that defines the behavior of game objects. Scripts are associated with game objects.

Programing language and IDE



Detail in the following textbooks





Devices and Implementation

Application of AR & AR

- Medical: simulation for teaching, researching and treatment. AR provides hidden information to the surgeon like rate of the heartbeat, state of the person's organ, blood pressure etc.
- Military: AR can be used to display the real battlefield scene.
- Marketing: Automotive industry had firstly used AR for various marketing purposes. Advertising fashion products and online smart business
- Education: simulation in training and teaching
- Assistance: virtual assistance, support in treatment
- Entertainment: Film, Game, Music and Movies
- ...
- https://smarttek.solutions/interesting-uses-of-vr-ar-technology/
- https://www.geeksforgeeks.org/top-7-modern-day-applications-of-augmented-reality-ar/
- https://www.liainfraservices.com/blog/top-5-applications-of-augmented-reality-ar/
- Lucio Tommaso et.al. Proceeding AR, VR, and Computer Graphics 2020.pdf

Example

- A guideline to create a VR application by using Oculus Quest on the MacOS
- See attached file.